

Resident and transient microbial taxa in the gill microbiome of the mussel *Mytilus galloprovincialis*

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Abstract

Microorganisms associate with marine bivalves, forming microbial communities collectively known as the bivalve's microbiome. These communities may include resident taxa, associated with the bivalve host throughout its life, and transient taxa, whose presence is temporary and attributed to the passage of, for example, seawater or food through the host. To assess the gill microbiome of the Mediterranean mussel, *Mytilus galloprovincialis* Lamarck, 1819, and its resident and transient taxa, individuals of this important aquaculture species were sampled at two locations on the eastern coast of the Adriatic Sea, Lim Bay and Mali Ston Bay, over one year. The microbiome was characterised using Illumina MiSeq sequencing of the V4 region of the 16S rRNA gene. For comparison, the microbial community in the surrounding seawater and sediment was also assessed at the same time and locations. The structure of the gill microbiome, determined at the OTU level, was distinct from the microbial community in both the surrounding seawater and sediment and did not differ between the two locations. However, it showed differences between months, indicating temporal dynamics. Taxonomic analysis revealed the presence of both resident and transient microbial taxa in the gill microbiome. Members without known relatives within *Endozoicomonadaceae* (i.e. unclassified *Endozoicomonadaceae*) were present in the gills of mussels from both locations and throughout the study period and appeared to be resident members. In contrast, *Endozoicomonas* and “*Candidatus* Endoecteinascidia” were transient, appearing in the gill microbiome of mussels from both locations only in certain months, as were *Vibrio* and members without known relatives within *Vibrionaceae* (i.e. unclassified *Vibrionaceae*), with the distinction that these were specific to Lim Bay. Overall, the results of this study show that the gill microbiome of the mussel *Mytilus galloprovincialis* differs from the microbial community in the surrounding seawater and sediment, exhibits temporal dynamics, and contains both resident and transient microbial taxa.

Keywords: gill microbiome, Mediterranean mussel, *Mytilus galloprovincialis*, aquaculture, Adriatic Sea, 16S rRNA gene sequencing, temporal changes, *Gammaproteobacteria*

26 Introduction

27 Marine animals, including bivalves, are associated with various microorganisms such as
28 archaea, bacteria, viruses, protists, and fungi [1, 2]. These microbes, collectively known as the
29 animal's microbiome, can reside on or within the animal's body [1, 3]. If located inside the body,
30 they may live either outside or within host cells. Notably, microbes within host cells can inhabit
31 specialised cells dedicated to housing symbionts, forming a particularly intimate type of symbiosis
32 [3]. The close relationship between the host and its microbiome can be viewed as a holobiont:
33 an integrated community in which the animal and its symbiotic microbes support each other [2,
34 4]. Furthermore, as suggested by the hologenome concept, the genetic content of the host and the
35 microbes in its microbiome should be considered a single evolutionary unit, meaning that host
36 evolution is influenced by its microbiome [2, 5].

37 Studies on bivalves, such as mussels, oysters, clams, and cockles, have shown that the bivalve's
38 microbiome may include both resident members, associated with the host throughout its life, and
39 transient members, whose presence is temporary and attributed, for example, to the passage of
40 seawater or food through the animal [6, 7]. Evidence from oysters suggests that most microbes
41 in the bivalve's microbiome are resident members [8]. Additionally, as previously reported for
42 oysters, these microbes may be acquired by the host vertically, transmitted from parent to offspring,
43 horizontally, adopted from the surrounding environment, or through a combination of these processes
44 [3, 9, 10]. Studies have also shown that the microbiome of various bivalves, such as mussels, oysters,
45 clams, and cockles, differs from the microbial community in surrounding seawater or sediment,
46 highlighting the resident nature of most microbiome members [6, 10–16]. In addition to differing
47 from the surrounding environment, the microbiome of different bivalve species also varies between
48 body parts, such as the gut, gills, and haemolymph [10, 13, 14, 16, 17], and is, to some extent and
49 for certain body parts, specific to the host species [6, 11, 15, 16, 18]. Although most microbes in the
50 bivalve's microbiome appear to be resident members, a portion remains transient [8], as suggested
51 by studies of mussels and oysters reporting temporal changes [11, 15, 19, 20]. The bivalve's

microbiome can also differ between geographic locations [8, 18, 19, 21, 22], indicating that local conditions may also have an influence. Interestingly, the microbiome of mussels transported from one geographic location to another converges towards that of native bivalves, highlighting the importance of local conditions [23, 24].

Microbes forming the bivalve's microbiome belong to various high-rank bacterial taxa, including the *Pseudomonadota* classes *Alphaproteobacteria* and *Gammaproteobacteria*, *Bacillota*, *Bacteroidota*, *Fusobacteriota*, and *Spirochaetota* [6, 18, 25]. Members of these taxa have also been found in the microbiome of the Mediterranean mussel, *Mytilus galloprovincialis* Lamarck, 1819 [6, 11, 14, 16, 21, 23, 26–31], an important aquaculture bivalve species widely distributed across the Mediterranean Sea, including the Adriatic Sea [32–36]. Studies focusing on *M. galloprovincialis* have reported patterns of dynamics for this species' microbiome similar to those previously mentioned for the bivalve's microbiome in general. Specifically, differences between the microbiome and the ambient seawater have been reported [11, 14, 16, 28, 31], as well as microbiome specificity for this host species [6, 11, 16] and its variation between body parts [14, 16], sampling periods [11, 23, 28, 30], and geographic locations [21, 23, 28].

In general, studies assessing the bivalve's microbiome have focused on gut tissue [2]. Similarly, research on the *M. galloprovincialis* microbiome has usually examined the gut microbiome [11, 16, 21, 28, 30] or assessed the microbiome of the whole individual without a tissue-specific sampling approach [6, 23, 27, 29]. Consequently, a limited number of studies have described the temporal and spatial dynamics of the microbiome in other body parts, such as gills and haemolymph [24, 26, 31]. As the mussel *M. galloprovincialis* is a filter-feeding animal [37], its gills form a uniquely complex interface between the host and the surrounding environment, are in constant contact with seawater, and may be influenced by sediment resuspension [38, 39]. Therefore, this study focused on assessing the gill microbiome of individuals of this bivalve species, as well as the microbial community in the surrounding seawater and sediment, from two important and geographically distant bivalve aquaculture locations on the eastern coast of the Adriatic Sea, namely Lim Bay and Mali Ston Bay

78 [40], over a period of one year. The study locations are in two legally protected marine reserve areas
79 with distinct hydrological conditions and productivity, typical of the Adriatic Sea biogeographical
80 regions to which they belong [41–44]. More precisely, the aims of this study were to: (i) compare
81 the mussel gill microbiome with the microbial community in the surrounding seawater and sediment,
82 (ii) determine differences in the gill microbiome between the two aquaculture locations, and (iii)
83 assess changes in the gill microbiome during the one-year study period. Unlike earlier work, this
84 study uses multiple biological replicates to capture inter-individual variation in the gill microbiome,
85 as this tissue is a dynamic barrier that continually interacts with its environment.

Methods

Sampling

This study was conducted at two bivalve aquaculture locations: Lim Bay in the north and Mali Ston Bay in the south of the eastern Adriatic Sea coast (Fig. 1). Mussels of the species *M. galloprovincialis*, seawater, and sediment were sampled in July, September, and November 2020 and January, March, and May 2021 (Supplementary Tables S1 and S2), as described in Žurga et al. [45] and Orlić et al. [46]. Briefly, locally produced adult mussels were deployed at an aquaculture site at a depth of 10 m in Lim Bay (45°7'50"N, 13°41'20"E) and 6 m in Mali Ston Bay (42°52'16"N, 17°42'12"E), following common farming practices at these locations (Fig. 1). Bottom depths in Lim Bay and Mali Ston Bay were 28 and 8 m, respectively. At each sampling date, 10 individuals were collected, their gills excised, and snap-frozen in liquid nitrogen. Seawater was collected at a depth close to that at which mussels were deployed, specifically 8 m in Lim Bay and 5 m in Mali Ston Bay, using a Niskin bottle and filtered (0.35–1 l) through 0.2 µm membrane filters. Filters were transferred into 2 ml cryotubes and snap-frozen in liquid nitrogen. As this is the initial study conducted in these coastal areas of the eastern Adriatic Sea, sediment samples were also analysed to establish a baseline characterisation of sediment microbial communities associated with bivalve farming environments. Sediment was also sampled because mussels constantly filter large volumes of seawater through their gills, and the seawater they filter may contain sediment particles resuspended by continuous seafloor disturbance caused by currents and waves [38, 39, 47, 48]. Surface sediment exposed to and disturbed by the movement of water masses was collected using an Ekman grab sampler, transferred into 2 ml cryotubes, and snap-frozen in liquid nitrogen. Upon arrival at the laboratory, samples of gills and sediment, as well as membrane filters, were stored at –80 °C until further processing. To provide a broader environmental context, seawater and sediment were sampled not only at the aquaculture site but also at a nearby site where no mussel-growing structures were present and which was sufficiently close to minimise spatial variability in baseline seawater and



Fig. 1 Sampling locations and sites in the Adriatic Sea. At each location, namely Lim Bay and Mali Ston Bay, samples were collected from one aquaculture site and one control site (© OpenStreetMap contributors, <https://www.openstreetmap.org/copyright>)

sediment characteristics. This additional site in both Lim Bay ($45^{\circ}7'57''\text{N}$, $13^{\circ}38'27''\text{E}$) and Mali Ston Bay ($42^{\circ}51'59''\text{N}$, $17^{\circ}42'5''\text{E}$; Fig. 1) was designated as the control site. Seawater at this site was sampled at the same depth as at the aquaculture site. Seawater abiotic parameters (temperature, salinity, dissolved oxygen, and chlorophyll *a*) displayed seasonal patterns typical of the eastern Adriatic Sea [45]. Sediment was classified as slightly gravelly mud at the aquaculture site and slightly gravelly sandy mud at the control site in Lim Bay, and as gravelly mud at the aquaculture site and gravelly muddy sand at the control site in Mali Ston Bay (unpublished data).

DNA isolation

Total DNA from gills (32–123 mg of snap-frozen tissue) and 0.2 µm membrane filters was isolated according to Massana et al. [49] with slight modifications. After extraction with organic solvents, 1/10 volume of 3 M sodium acetate (pH 5.2) was added. DNA was precipitated by adding 1 volume of cold (–20 °C) isopropanol and incubating the mixture for 1 h or overnight at –20 °C. The pellet obtained after centrifugation at 20,000 × g and 4 °C for 20 min was washed twice with 500 µl of cold (–20 °C) 70% ethanol. After each wash, the pellet was centrifuged at 20,000 × g and 4 °C for 5 min. Air-dried pellets were resuspended in 100 µl of deionised water for DNA isolated from gills and 50 µl for DNA isolated from membrane filters. Total DNA from sediment was isolated as described in Markovski et al. [50]. This is a modified protocol [51] of the isolation procedure described in Zhou et al. [52]. Briefly, 2 g of sediment was mixed with extraction buffer and proteinase K, and incubated by horizontal shaking at 37 °C for 30 min. After SDS was added, the mixture was incubated again by horizontal shaking at 65 °C for 60 min. Sediment particles were removed by centrifugation and the supernatant was extracted three times with an equal volume of chloroform:isoamyl alcohol (1:1). DNA was precipitated by adding isopropanol and incubating the mixture at 22 °C for 60 min. The pellet obtained by centrifugation was washed twice with cold (–20 °C) 70% ethanol, air-dried, and resuspended in 100 µl of deionised water. DNA isolated from gills, membrane filters, and sediment was treated with RNase A at a final concentration of 200 µg ml^{–1} at 37 °C for 2 h. DNA concentration was determined using the Quant-iT PicoGreen dsDNA Assay Kit (Invitrogen, USA) according to the manufacturer's instructions. DNA from gills was diluted to 10 ng µl^{–1}, and DNA from membrane filters and sediment to 1 ng µl^{–1}.

Illumina 16S rRNA sequencing

The V4 region of the 16S rRNA gene was amplified using a two-step PCR procedure. In the first PCR, 515F (Parada; 5'-GTGYCAGCMGCCGCGGTAA-3') and 806R (Apprill;

5'-GGACTACNVGGGTWTCTAAT-3') primers from the Earth Microbiome Project (<https://earthmicrobiome.org/protocols-and-standards/16s>) were used [53–56], each with a sequence tag at the 5' end. Four parallel PCRs were performed to amplify the target region from each DNA sample. Each 25 µl reaction contained 1 × Q5 Reaction Buffer, 0.2 mM dNTP mix, 0.7 mg ml⁻¹ bovine serum albumin (BSA), 0.5 µM forward and reverse primers, 0.5 U Q5 High-Fidelity DNA Polymerase (New England Biolabs, USA), and 50 ng DNA template from gills or 5 ng from membrane filters and sediment. The cycling conditions were: initial denaturation at 98 °C for 30 s; 25 cycles of denaturation at 98 °C for 10 s, annealing at 50 °C for 30 s, and elongation at 72 °C for 5 s for reactions with DNA from gills or 30 s for reactions with DNA from membrane filters and sediment. PCR cycling ended with a final elongation at 72 °C for 2 min. The higher amount of DNA template (50 ng) and shorter elongation time (5 s) used for amplifying the target region from DNA isolated from gills were optimised to reduce amplification of non-specific fragments. The four parallel reactions were pooled, and PCR products were purified using the GeneJET PCR Purification Kit (Thermo Fisher Scientific, USA) according to the manufacturer's instructions. The protocol without isopropanol addition was followed to avoid co-purification of primer-dimers. The column was eluted in 30 µl deionised water.

Purified PCR products were sent to IMGM Laboratories, Martinsried, Germany, for Illumina MiSeq sequencing (2 × 250 bp). Before sequencing, the second PCR in the two-step PCR procedure was performed using primers that bind to the sequence tag added during the first PCR. In addition to the gill, seawater, and sediment samples, two positive and two negative controls were sequenced. Each positive control consisted of four parallel PCRs using genomic DNA from a mock community composed of evenly mixed DNA from 20 bacterial strains (ATCC MSA-1002, ATCC, USA) as the DNA template. Each negative control comprised four parallel PCRs without a DNA template. One positive and one negative control were processed according to the PCR procedure for gill samples, and one positive and one negative control according to the PCR procedure for seawater and sediment samples.

Sequence and data analysis

Sequences obtained were analysed using mothur (version 1.48.2) [57] following the MiSeq Standard Operating Procedure (MiSeq SOP; https://mothur.org/wiki/miseq_sop) [58] and recommendations from the Riffomonas project to enhance data reproducibility (<https://riffomonas.org>). For alignment and classification, the SILVA SSU Ref NR 99 database (release 138.2; <https://www.arb-silva.de>) was used [59, 60]. Sequences were clustered into operational taxonomic units (OTUs) using a 97% similarity threshold. In this study, sequences that could not be classified into any taxon were labelled “No Relative”, whereas sequences that could not be classified to any known taxon within a specific taxonomic group are referred to as members without known relatives within that group.

Data processing and visualisation were performed using R (version 4.6.1) [61] with the packages vegan (version 2.7-5) [62] and tidyverse (version 2.0.0) [63, 64], as well as several other packages [65–88]. Mean values of the observed number of OTUs, Chao1, abundance-based coverage estimator (ACE), exponential of the Shannon diversity index, and Inverse Simpson diversity index were calculated from 100 subsamples of OTU abundances (rarefaction) to account for differences in sequencing depth [89]. Each subsample was obtained by randomly selecting, without replacement, a number of sequences equal to the minimum number of sequences per sample using vegan’s function `rrarefy` [62]. The richness estimators Chao1 and ACE were obtained using vegan’s function `estimateR`, while the Shannon and Inverse Simpson diversity indices were calculated using vegan’s function `diversity` [62]. To express both diversity indices as the effective number of OTUs, the exponential of the Shannon diversity index was calculated [90]. Non-metric multidimensional scaling (NMDS) was performed on Bray–Curtis dissimilarities using vegan’s function `metaMDS` [62, 91, 92]. To account for differences in sequencing depth, Bray–Curtis dissimilarities were calculated using vegan’s function `avgdist` by randomly subsampling OTU abundances to the minimum number of sequences per sample 100 times and calculating mean Bray–Curtis dissimilarities across all subsamples (rarefaction) [62, 89]. Differences between communities from distinct

environments, sites, locations, and months were tested by performing permutational multivariate analysis of variance (PERMANOVA) on the same Bray–Curtis dissimilarities using `vegan`’s function `adonis2` with 999 permutations, followed by pairwise PERMANOVA comparisons using `pairwiseAdonis`’s function `pairwise.adonis`, also with 999 permutations, when more than two groups of samples were tested (Bonferroni correction was applied to address the multiple comparisons problem) [62, 76]. Multivariate homogeneity of group dispersions was assessed using `vegan`’s function `betadisper` and tested using `vegan`’s function `permutest` with 999 permutations [62]. Significant PERMANOVA results accompanied by significant differences in the homogeneity of group dispersions ($p < 0.05$) were taken into account when group sizes were balanced or, in one case, reported separately [93]. Differences between richness estimators, diversity indices, and proportions of taxa were tested using the Mann–Whitney U test with the function `wilcox.test` when two groups were compared, or the Kruskal–Wallis H test with the function `kruskal.test`, followed by pairwise comparisons with the Mann–Whitney U test using the function `pairwise.wilcox.test`, when more than two groups were compared [61]. The Bonferroni correction was applied to address the multiple comparisons problem.

From six sampling months, two locations (Lim Bay and Mali Ston Bay), and two sites (aquaculture and control), 24 seawater and 24 sediment samples were processed. Additionally, from 10 individuals collected at the aquaculture site during the same months and locations, 120 gill samples were processed. This resulted in a total of 168 samples, of which 149 yielded sufficient sequences for further analysis (Supplementary Tables S1 and S2). From these samples, a total of 2.1 million sequences were obtained after quality curation and exclusion of sequences without known relatives (no relative sequences), as well as eukaryotic, chloroplast, and mitochondrial sequences. The number of sequences per sample ranged from 2,062 to 60,611 (Supplementary Tables S1 and S2). These sequences were clustered into a total of 66,695 different OTUs. For seawater and sediment samples, as well as for gill samples with a high number of OTUs, the rarefaction curves did not level off, even with the highest sequencing effort, as is commonly observed in high-throughput 16S rRNA amplicon sequencing approaches (Supplementary Figs. S1 and S2). However, for gill

221 samples with a low number of OTUs, the rarefaction curves did level off. The absence of levelling
222 off in the rarefaction curves for seawater, sediment, and some gill samples suggests that increasing
223 sequencing depth could reveal additional OTUs. Based on the positive controls (mock community
224 ATCC MSA-1002), a sequencing error rate of 0.04% was calculated, which is comparable to
225 previously reported values for next-generation 16S rRNA amplicon sequencing [50, 58, 94, 95]. In
226 addition, all 20 strains whose DNA was present in the mock community were detected. The two
227 negative controls processed with the samples yielded 10 and 12 sequences after sequence quality
228 curation.

Results

To assess the richness and diversity of microbial communities, the observed number of OTUs, the richness estimators Chao1 and ACE, the exponential of the Shannon diversity index, and the Inverse Simpson diversity index were calculated at the OTU level for the sampled environments (seawater, sediment, and the gills of *M. galloprovincialis*) at both aquaculture and control sites (Fig. 2). Statistically significant changes were found for all parameters (Kruskal–Wallis H test, all $p < 0.0001$; Supplementary Table S3). The sediment microbial community had the highest observed number of OTUs (aquaculture, 928.9–1,343.4 OTUs; control, 1,049.7–1,448.1 OTUs), followed by the seawater community (aquaculture, 188.4–423.8 OTUs; control, 216.7–455.9 OTUs), and the gill microbiome (aquaculture, 39.8–746.9 OTUs). No significant differences in the observed number of OTUs were found between the aquaculture and control sites for the sediment and seawater communities (Fig. 2 and Supplementary Table S3). The same pattern was observed for the exponential of the Shannon diversity index and the Inverse Simpson diversity index. However, when the richness estimators Chao1 and ACE were calculated, differences between the gill microbiome and seawater microbial communities at the aquaculture and control sites were no longer observed (Fig. 2 and Supplementary Table S3). Additionally, location-specific changes for all sampled environments, as well as temporal changes for the gill microbiome, were evaluated. Neither parameter showed differences between the sampling locations, except for the Inverse Simpson diversity index of sediment communities, which showed higher values in Mali Ston Bay (175.3–803.1 OTUs) than in Lim Bay (106.4–289.6 OTUs; Mann–Whitney U test, $p < 0.01$). Furthermore, for the gill microbiome, only the richness estimators Chao1 and ACE showed differences across sampling months (Kruskal–Wallis H test, both $p < 0.05$). However, comparisons between sampling months using the pairwise Mann–Whitney U test showed no significant differences (Bonferroni-corrected).

To assess variation in the structure of microbial communities, Bray–Curtis dissimilarities based on OTU abundances were calculated and analysed using NMDS (Fig. 3). When all samples were

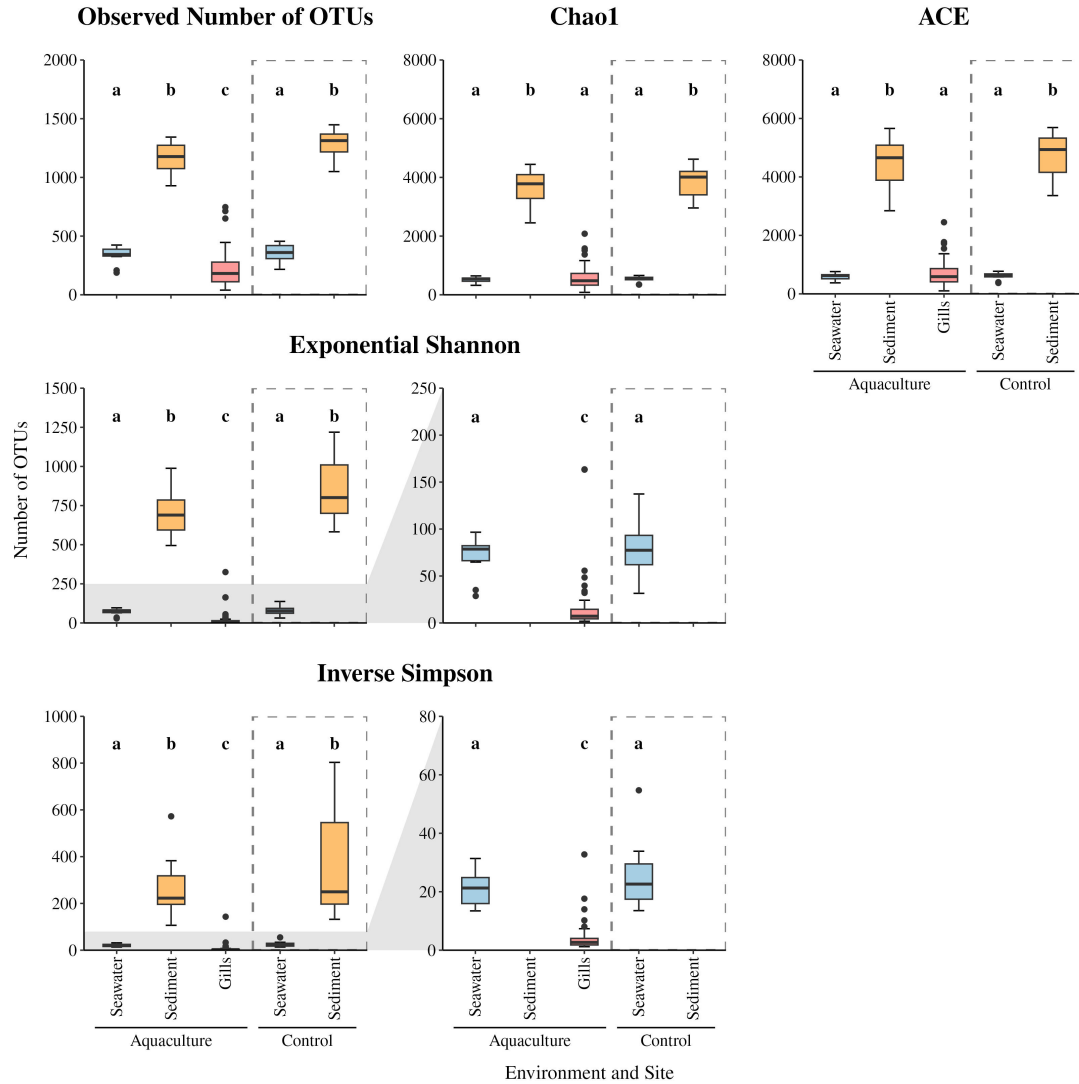


Fig. 2 The observed number of OTUs, Chao1, ACE, exponential of the Shannon diversity index, and Inverse Simpson diversity index of microbial communities from different environments and sites. Dashed boxes indicate groups of samples from the control site, while non-dashed areas represent groups of samples from the aquaculture site. Groups that do not share a letter are significantly different ($p < 0.05$). Diagrams representing the exponential of the Shannon diversity index and the Inverse Simpson diversity index consist of a main panel showing the complete dataset and a magnified panel illustrating more clearly the differences between the microbial communities from seawater and from the gills of the mussel *M. galloprovincialis* (gill microbiome)

analysed together, a pronounced and statistically significant distinction between communities from different environments was observed (PERMANOVA, $R^2 = 0.35$, $p < 0.01$, significant differences in the homogeneity of group dispersions [$p < 0.05$] with unequal group sizes). Pairwise comparisons

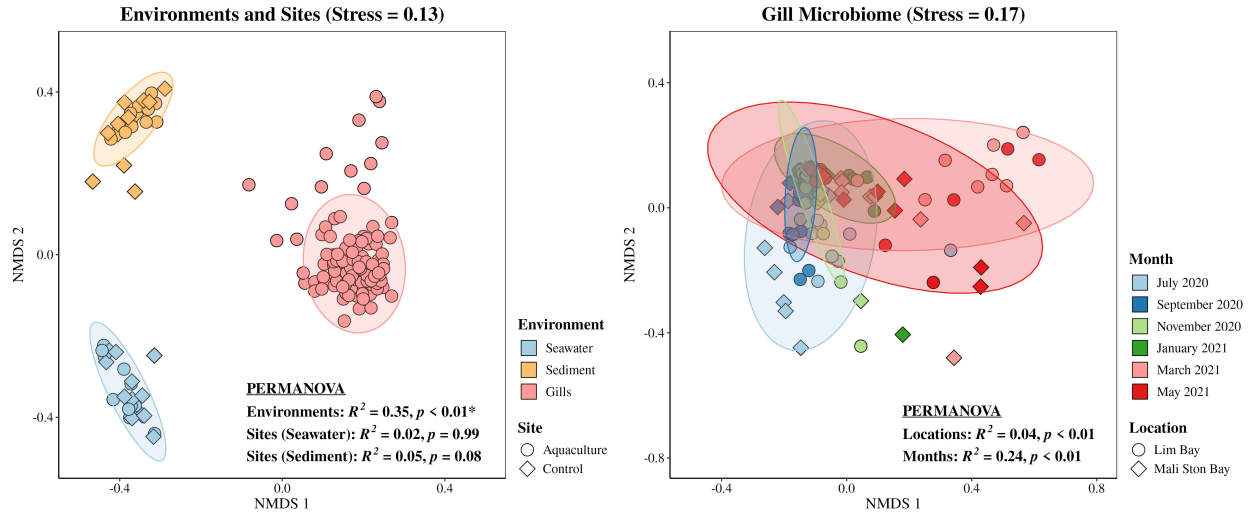


Fig. 3 NMDS of Bray–Curtis dissimilarities calculated from OTU abundances in microbial communities from different environments and sites, and in microbial communities from the gills of the mussel *M. galloprovincialis* (gill microbiome) sampled in different months in both Lim Bay and Mali Ston Bay. The stress value for each NMDS is given in the corresponding plot title in parentheses. Ellipses represent 95% confidence intervals for each environment and sampling month. PERMANOVA test results for differences between microbial communities from different environments, sites, sampling months, and locations are shown in the corresponding plots. *significant differences in the homogeneity of group dispersions ($p < 0.05$) with unequal group sizes

also showed significant differentiation among all environments (pairwise PERMANOVA, $R^2 = 0.25$ – 0.30 , all $p < 0.01$, Bonferroni-corrected). In contrast, there was no significant distinction in seawater (PERMANOVA, $R^2 = 0.02$, $p = 0.99$) or sediment (PERMANOVA, $R^2 = 0.05$, $p = 0.08$) communities between aquaculture and control sites (Fig. 3). When the influence of sampling location, Lim Bay and Mali Ston Bay, was assessed, sediment microbial communities exhibited significant location-specific differentiation, with some community overlap (PERMANOVA, $R^2 = 0.11$, $p < 0.01$). In contrast, seawater microbial communities did not show significant differentiation between locations (PERMANOVA, $R^2 = 0.06$, $p = 0.17$), while location-specific differentiation of gill microbiomes was significant but almost negligible (PERMANOVA, $R^2 = 0.04$, $p < 0.01$; Fig. 3). When the influence of sampling month on the gill microbiome was assessed, significant temporal differentiation was observed (PERMANOVA, $R^2 = 0.24$, $p < 0.01$, samples from January 2021 were excluded to avoid unequal group sizes; Fig. 3). Pairwise comparisons showed significant differentiation among all months (pairwise PERMANOVA, $R^2 = 0.12$ – 0.29 , all $p <$

0.05, Bonferroni-corrected, samples from January 2021 were excluded to avoid unequal group sizes), except between September and November 2020 (pairwise PERMANOVA, $R^2 = 0.05$, $p = 0.61$, Bonferroni-corrected) and between March and May 2021 (pairwise PERMANOVA, $R^2 = 0.04$, $p = 1.00$, Bonferroni-corrected).

To compare the taxonomic composition of microbial communities between the sampled environments and sites, the average relative contribution of each taxon across samples from both locations and all sampling months was calculated (Fig. 4). Differences in community structure between environments and similarities between sites identified by the NMDS were also evident in the taxonomic composition of the microbial communities. There were no notable differences between the aquaculture and control sites for the sediment and seawater communities. The seawater microbial community was mainly composed of the bacterial taxa *Alphaproteobacteria* (aquaculture, 12.8–39.0%; control, 11.1–42.1%), *Gammaproteobacteria* (aquaculture, 14.6–34.6%; control, 13.4–32.7%), *Bacteroidota* (aquaculture, 7.5–19.3%; control, 8.4–20.3%), *Cyanobacteriota* (aquaculture, 1.0–23.2%; control, 1.2–21.6%), *Verrucomicrobiota* (aquaculture, 0.7–25.9%; control, 0.8–24.5%), and the archaeal phylum *Thermoplasmatota* (aquaculture, 0.01–32.2%; control, 0–32.5%). The sediment microbial community differed from the seawater community and was mainly composed of the bacterial taxa *Gammaproteobacteria* (aquaculture, 12.6–22.5%; control, 11.1–24.7%), *Thermodesulfobacteriota* (aquaculture, 8.3–16.7%; control, 2.0–16.6%), *Planctomycetota* (aquaculture, 9.8–15.3%; control, 7.4–15.2%), *Bacteroidota* (aquaculture, 4.1–11.1%; control, 2.9–15.8%), *Chloroflexota* (aquaculture, 3.3–7.2%; control, 1.3–7.0%), members without known relatives within *Bacteria* (aquaculture, 2.5–13.1%; control, 3.7–18.0%), and the archaeal phylum *Thermoproteota* (aquaculture, 1.6–13.1%; control, 0.5–12.0%; Fig. 4). In contrast to the taxonomic composition of both the seawater and sediment microbial communities, the gill microbiome was largely composed of the bacterial class *Gammaproteobacteria* (aquaculture, 27.0–98.8%), while the proportions of other high-rank bacterial groups, such as *Alphaproteobacteria* (aquaculture, 0.2–33.7%), *Bacteroidota* (aquaculture, 0.2–50.4%), *Planctomycetota* (aquaculture, 0.02–8.3%), *Bacillota* (aquaculture, 0.03–26.2%) or members without known relatives within

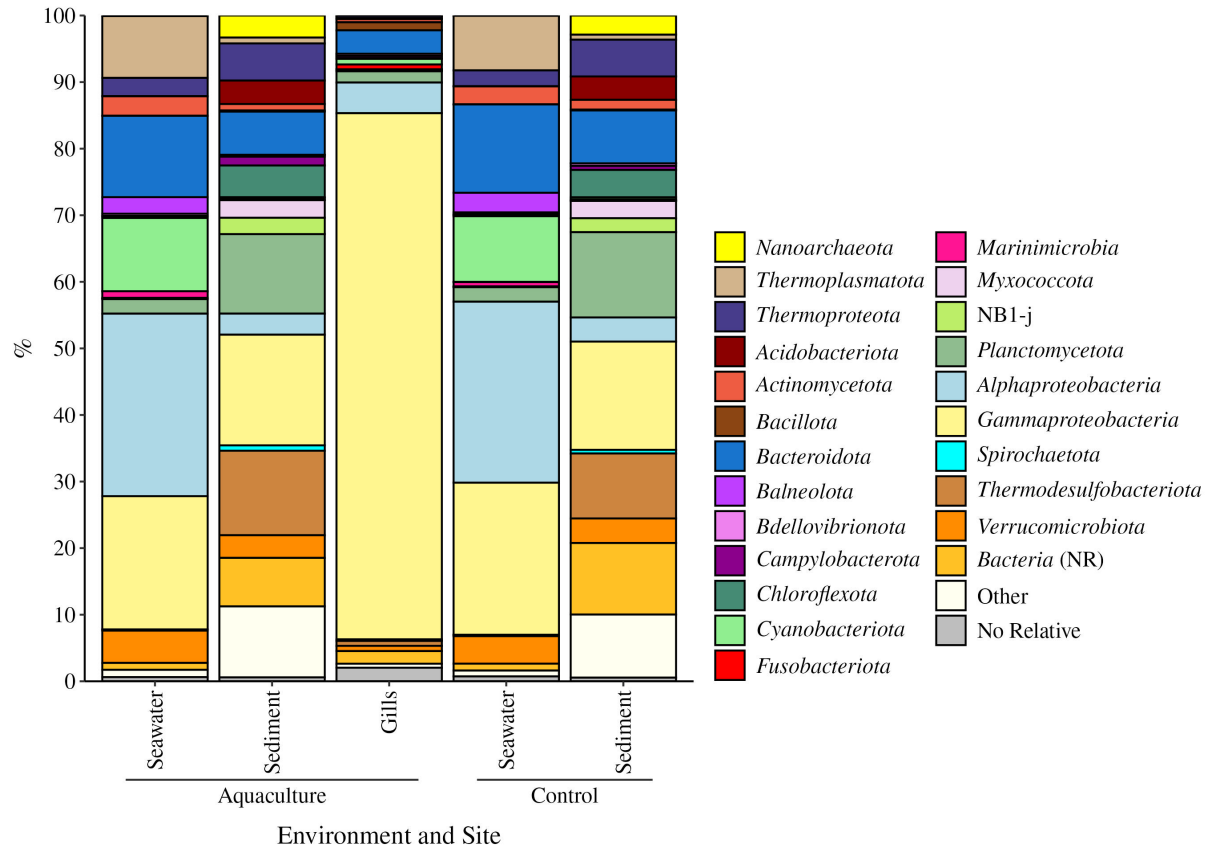


Fig. 4 Taxonomic classification and average relative contribution of the most abundant ($\geq 3\%$) bacterial and archaeal sequences in different sampled environments and at each site. The average relative contribution of each taxon across samples from both Lim Bay and Mali Ston Bay, and across all sampling months, was calculated. “No Relative” indicates sequences that could not be classified into any taxon, while “NR” (No Relative) next to the name of a taxonomic group refers to sequences that could not be assigned to any known taxon within that group

Bacteria (aquaculture, 0.1–13.1%), were substantially lower (Fig. 4). The high proportion of *Gammaproteobacteria* in the gill microbiome was observed in individuals at both locations and throughout the study period (Fig. 5), while *Alphaproteobacteria*, *Bacteroidota*, and *Bacillota* comprised a notable proportion of the gill microbiome only in a few individuals (Supplementary Fig. S4).

A more detailed analysis of the taxonomic composition of the seawater microbial community revealed temporal patterns for some taxa (Supplementary Fig. S3). For example, the archaeal phylum *Thermoplasmatota* showed a similar pattern at both locations and sites, comprising a

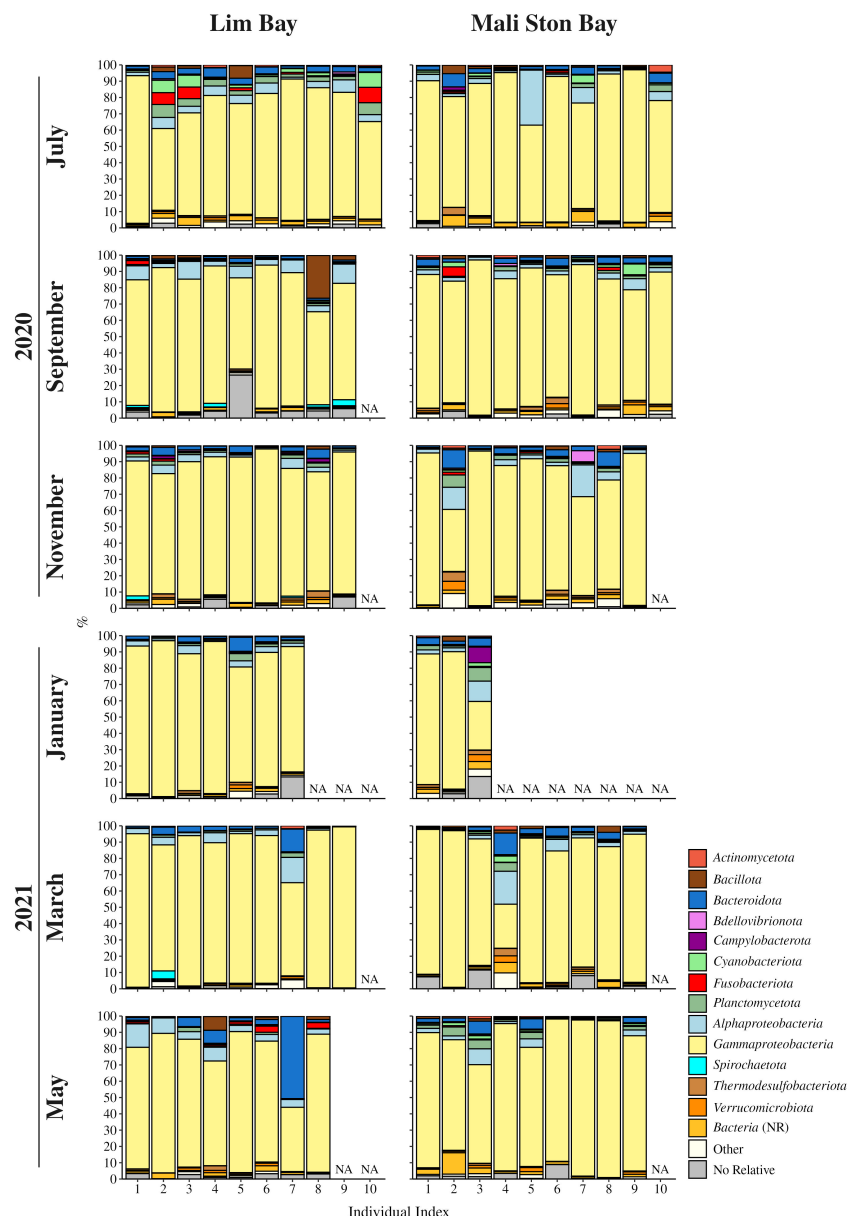


Fig. 5 Taxonomic classification and relative contribution of the most abundant ($\geq 3\%$) bacterial and archaeal sequences from the gills of the mussel *M. galloprovincialis* (gill microbiome). From July 2020 to May 2021, 10 mussels were collected per sampling month in Lim Bay and Mali Ston Bay. Each bar (individual index) represents an individual mussel (see Supplementary Table S2). “No Relative” indicates sequences that could not be classified into any taxon, while “NR” (No Relative) next to the name of a taxonomic group refers to sequences that could not be assigned to any known taxon within that group. “NA” (Not Available) indicates samples that were not successfully analysed

higher proportion of the community from November 2020 to January 2021 (10.9–32.5%) than in other months (0–4.6%). Notably, during the months when the proportion of this archaeal

phylum was high, almost the entire *Thermoplasmatota* community consisted of Marine Group II (95.2–100%). Marked differences between locations were observed in May 2021, when in Lim Bay at both sites, more than half of the microbial community was composed of *Gammaproteobacteria* (31.5–34.6%) and *Verrucomicrobiota* (24.5–25.9%), which was higher than in Mali Ston Bay at the same time (*Gammaproteobacteria*, 20.2–23.6%; *Verrucomicrobiota*, 3.9–3.9%) or in other months (*Gammaproteobacteria*, 13.4–32.7%; *Verrucomicrobiota*, 0.7–4.7%). In contrast, *Alphaproteobacteria* constituted a lower proportion of the microbial community in Lim Bay in May 2021 (11.1–12.8%) than in Mali Ston Bay at the same time (31.0–39.0%) or in other months (19.6–42.1%). For the sediment microbial community, a fairly similar taxonomic composition between locations and sites, as well as across sampling months, was observed (Supplementary Fig. S3). The only notable exception was the sediment community in Mali Ston Bay at the control site, where, for example, *Thermodesulfobacteriota* comprised a lower proportion of the community from July to November 2020 (2.0–5.1%) than from January to May 2021 (10.8–16.6%). This low proportion of *Thermodesulfobacteriota* was also lower than that observed at the aquaculture site in Mali Ston Bay (9.9–16.7%) and at both sites in Lim Bay (8.3–15.2%).

Because, as previously mentioned, the gill microbiome of most *M. galloprovincialis* mussels, except for a few individuals, was largely composed of *Gammaproteobacteria*, no differences in community composition between locations or sampling months were identified (Fig. 5). However, when the taxonomic composition was analysed at lower taxonomic ranks, such as family and genus, differences between locations and sampling months were observed (Fig. 6). Although high variability in the taxonomic composition of each individual's gill microbiome was noted, low-rank gammaproteobacterial groups present at both locations and throughout the study period, as well as groups specific to certain locations or months, were identified. In addition to these patterns, samples that did not yield sufficient sequences for further analysis were also specific to a particular month, January 2021, when 10 samples did not yield sufficient sequences, whereas in all other months only eight samples lacked enough sequences for further analysis (Figs. 5 and 6). Members without known relatives within *Endozoicomonadaceae* were present in mussel gills at both

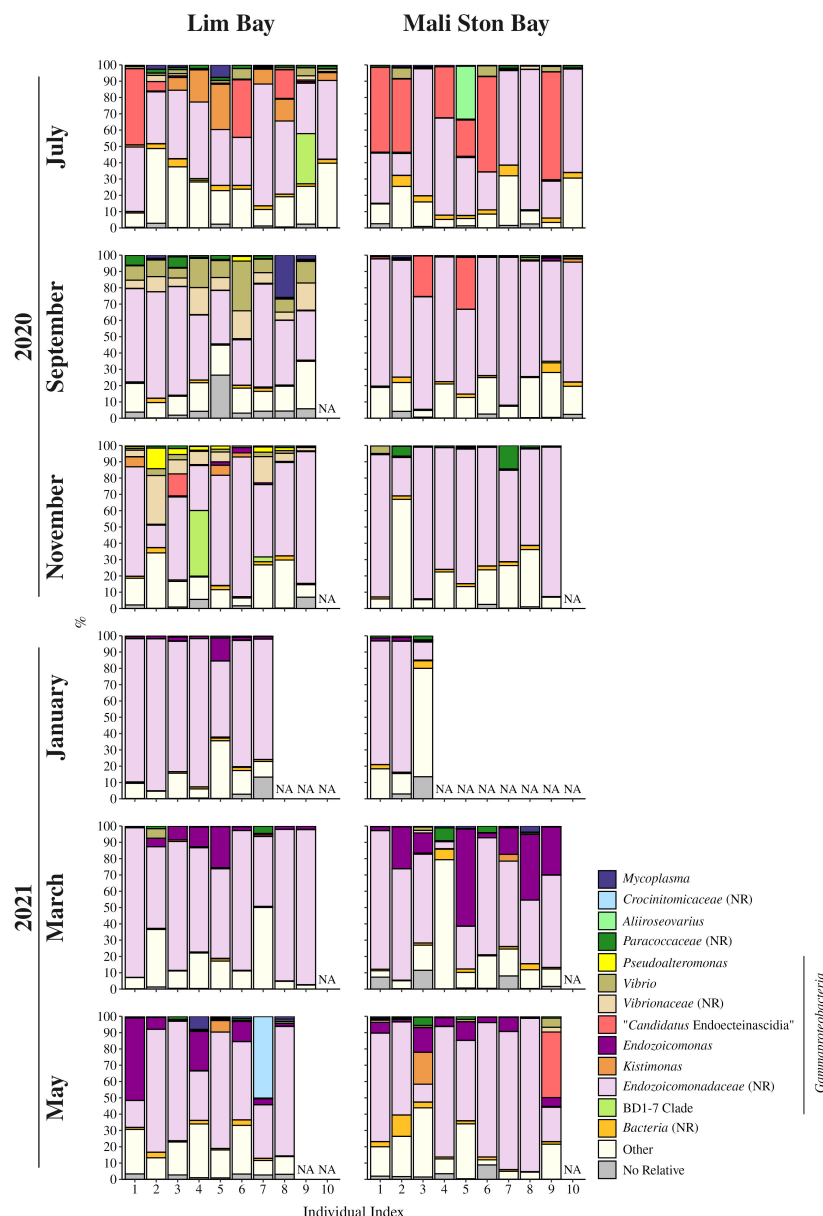


Fig. 6 Relative contribution of the most abundant ($\geq 10\%$) bacterial and archaeal sequences from the gills of the mussel *M. galloprovincialis* (gill microbiome) classified at the genus taxonomic rank. From July 2020 to May 2021, 10 mussels were collected per sampling month in Lim Bay and Mali Ston Bay. Each bar (individual index) represents an individual mussel (see Supplementary Table S2). “No Relative” indicates sequences that could not be classified into any taxon, while “NR” (No Relative) next to the name of a taxonomic group refers to sequences that could not be assigned to any known taxon within that group. “NA” (Not Available) indicates samples that were not successfully analysed

locations and throughout the study period, showing a high degree of variability between individuals (4.3–95.1%; Fig. 6). When differences in the proportion of this group between individuals from

different locations were tested, the only significant difference was its higher proportion in the gill microbiome of individuals from Mali Ston Bay (52.0–90.7%) compared to those from Lim Bay (27.9–66.6%) in September 2020 (Mann–Whitney U test, $p < 0.001$, testing for January 2021 was omitted because of the limited number of samples). In addition, significant differences between months were observed only in Lim Bay (Kruskal–Wallis H test, $p < 0.01$, samples from Mali Ston Bay from January 2021 were excluded due to the limited number of samples), specifically between July 2020 (29.4–74.7%) and January 2021 (46.7–93.2%), July 2020 and March 2021 (42.9–95.1%), and between September 2020 (27.9–66.6%) and January 2021 (pairwise Mann–Whitney U test, all stated comparisons $p < 0.05$, Bonferroni-corrected). Interestingly, *Endozoicomonas*, a member of the family *Endozoicomonadaceae*, was almost absent in the gill microbiome at both locations from July to November 2020 (0–3.2%), but from January to May 2021 it comprised a substantial proportion of the microbiome in many individuals from both locations (0.5–59.5%; Mann–Whitney U test, $p < 0.0001$; Fig. 6). Another taxonomic group showing specificity to a certain period was “*Candidatus* Endoecteinascidia”, which comprised a substantial proportion of the gill microbiome in many individuals in July 2020 (0–66.1%), while in other months the microbiomes of only a few individuals contained a higher proportion of this group (0–40.3%; Fig. 6). These differences in proportions of “*Candidatus* Endoecteinascidia” between July 2020 and all other months were supported by the Mann–Whitney U test (all $p < 0.01$, Bonferroni-corrected). *Vibrio* showed differences in the proportion of the gill microbiome between locations and sampling months (Fig. 6). In September 2020, higher proportions of this group were observed in gill microbiomes from Lim Bay (6.0–30.5%) than from Mali Ston Bay (0.1–1.5%). In addition, these proportions in September 2020 were also higher than those observed in all other months at either location (0–6.5%). These differences in the proportion of the gill microbiome between individuals from Lim Bay in September 2020 and individuals from all other months at either location were supported by the Mann–Whitney U test (all $p < 0.01$, Bonferroni-corrected, samples from Mali Ston Bay from January 2021 were excluded due to the limited number of samples). Furthermore, members without known relatives within *Vibrionaceae* also showed differences in the proportion of the gill microbiome between

locations and sampling months (Fig. 6). Higher proportions of this group were observed in individuals from Lim Bay in September (4.9–17.1%) and November (1.0–29.9%) 2020 than in individuals from Mali Ston Bay in the same months (September 2020, 0.008–0.3%; November 2020, 0.06–0.5%). In addition, these proportions in individuals from Lim Bay from September to November 2020 were also higher than those observed in individuals from all other months at either location (0–3.8%). The similarities in proportions of this group in individuals from Lim Bay in September and November 2020 and the differences from individuals from all other months at either location were supported by the Mann–Whitney U test (all $p < 0.01$, Bonferroni-corrected, samples from Mali Ston Bay from January 2021 were excluded due to the limited number of samples). Additionally, the relationship between members without known relatives within *Vibrionaceae* and *Vibrio* was assessed in individuals from Lim Bay in September 2020, when both groups showed higher proportions (Fig. 6). This assessment indicated that members without known relatives within *Vibrionaceae* accounted for 36.0–55.7% of their combined contribution.

The presence of taxa abundant in the gill microbiome (members without known relatives within *Endozoicomonadaceae*, *Endozoicomonas*, “*Candidatus* Endoecteinascidia”, *Vibrio*, and members without known relatives within *Vibrionaceae*) was also assessed in seawater (23 samples in total) and sediment (24 samples in total) microbial communities to evaluate the influence of microbes from these surrounding environments on the gill microbiome. Members without known relatives within *Endozoicomonadaceae* were detected in only three seawater and four sediment samples, and at very low proportions (seawater, 0.02–0.03%; sediment, 0.007–0.06%). *Endozoicomonas* was detected in nine sediment samples, again at very low proportions (0.005–0.1%), whereas “*Candidatus* Endoecteinascidia” was observed only in five seawater samples, also at very low proportions (0.03–0.2%). *Vibrio* and members without known relatives within *Vibrionaceae* were detected in more seawater and sediment samples. *Vibrio* was present in 10 seawater and 14 sediment samples, at proportions of 0.03–0.7% in the seawater community and 0.005–1.3% in the sediment community. Similarly, members without known relatives within *Vibrionaceae* occurred in 13 seawater and eight sediment samples, with proportions of 0.03–2.0% in the seawater community

391 and 0.008–0.2% in the sediment community. The presence or absence of these taxa in seawater and
392 sediment samples from specific locations or sampling months could not be associated with their
393 high proportions in the gill microbiome at the same locations and months.

Discussion

The mussel *M. galloprovincialis* is a filter-feeding animal, meaning that, as in most bivalve species, its gills are used for both gas exchange and feeding [37]. More specifically, it is a suspension-feeding bivalve that relies on the selective uptake of seston, comprising living and non-living material across a wide size range, to acquire suspended particles used as a food source, such as phytoplankton, microzooplankton, microbes, and detritus [38, 39, 96]. The flow of seawater required for feeding is generated by lateral cilia on the gills. Seawater flows through the inhalant aperture, passes through the interfilamentary spaces of the gills into the suprabranchial cavity, and exits through the exhalant aperture [38, 39]. Food particles are retained by the gills whose capture efficiency depends on particle size, with smaller particles, such as free-living microbes, being less likely to be retained than larger ones, including particle-attached microbes. Captured particles are then selectively processed by mucociliary mechanisms and are either ingested or rejected as pseudofaeces [38, 39, 97, 98]. As the gills, which are essential for this suspension-feeding mechanism, must be in relatively constant contact with seawater and may also be influenced by material resuspended from the seafloor [38, 39, 47, 48], it is therefore appropriate to compare the gill microbiome with the microbial communities in the surrounding seawater and sediment.

In this study, the observed number of OTUs, as well as the exponential of the Shannon diversity index and the Inverse Simpson diversity index, were significantly lower for the gill microbiome than for the seawater and sediment microbial communities. However, when the total number of OTUs is considered using calculated richness estimators, the difference in the number of OTUs between the gill microbiome and the seawater microbial community is no longer evident. This lower observed number of OTUs in the gill microbiome compared with the seawater microbial community, together with the absence of differences in the predicted number of OTUs based on richness estimators, is likely due to the mussel selectively excluding many rare seawater OTUs from its gill microbiome. A study on the gill microbiome of the same species reported differences in alpha diversity between the gill microbiome and the microbial community in the surrounding

seawater [31], while another study found no differences between the microbial community in the surrounding seawater and the microbiome of various body parts of this species, including the gills [14]. In any case, the alpha diversity of the gill microbiome and seawater microbial community was substantially lower than that of the sediment community. Similarly, comparisons of the gut and shell microbiome of the eastern oyster, *Crassostrea virginica*, with the sediment microbial community also found lower alpha diversity in the oyster's gut and shell microbiome than in the sediment community [12]. The alpha diversity of the gill microbiome did not differ between the two locations, Lim Bay and Mali Ston Bay, consistent with a study comparing the gill microbiome of *M. galloprovincialis* between nearby locations [31], as well as another study comparing its whole microbiome between more distant geographic locations [23]. Richness estimators were also the only alpha diversity parameters showing significant overall differences in the gill microbiome between sampling months, although pairwise comparisons did not show any significant differences between individual months. Overall, it appears that the mussel *M. galloprovincialis* has a gill microbiome with lower alpha diversity than the surrounding sediment microbial community, a lower observed number of OTUs and lower diversity indices than the community in the surrounding seawater, and no differences in predicted richness compared with the surrounding seawater community. In addition, its gill microbiome appears to be stable in terms of alpha diversity, showing minimal variation between different sampling locations and months.

Although richness estimators did not indicate differences between the gill microbiome of *M. galloprovincialis* and the microbial community in the surrounding seawater, NMDS performed on Bray–Curtis dissimilarity calculated from OTU abundances revealed a clear separation between the gill microbiome and the communities in both the surrounding seawater and sediment. This finding is consistent with studies reporting differences between the microbiome of this species' various body parts, such as the gut, gills, and haemolymph, and the community in the surrounding seawater [11, 14, 16, 28, 31], as well as with the distinct community structures observed between the gut and shell microbiomes of the eastern oyster and the microbial community in the surrounding sediment [12]. In addition, the high similarity of sediment microbial communities between the

aquaculture and control sites suggests low spatial heterogeneity in sediment properties over short physical distances from mussel-growing structures. However, while local impacts were not detected at the small scale studied, a larger-scale study across a wider area may be needed to fully evaluate potential interactions between farmed bivalves with the benthic environment [99]. Comparison of the overall structure of the *M. galloprovincialis* gill microbiome at the OTU level in this study did not reveal significant differentiation between the two locations, Lim Bay and Mali Ston Bay, despite their considerable geographic distance. This agrees with a study of the gill microbiome of the same species that also did not show variation between locations [31]. In contrast, the structure of the gut microbiome of the same species varied with geographic location [21, 28], indicating a greater influence of local conditions on the gut than on the gill microbiome, possibly due to variation in the origin of food, which also includes microbes, with geographic location and the potential involvement of the gut microbiota in food digestion [7, 96]. Interestingly, the structure of the overall microbiome of the mussel *M. galloprovincialis* transported from one geographic location to another is initially different and converges towards that of native mussels, highlighting the influence of local conditions, the possible high contribution of the gut microbiota to the overall microbiome, and the importance of assessing tissue-specific microbiomes [23]. Furthermore, in this study, the structure of the gill microbiome at the OTU level showed a separation of microbiomes from different sampling months, except between September and November 2020 and between March and May 2021, indicating the presence of microbiome taxa with temporal changes. Similarly, studies on the overall microbiome of the mussel *M. galloprovincialis* [23], as well as on the gut microbiome [11, 28, 30], also indicated temporal microbiome changes. In contrast, a study of the gill microbiome of the same species did not reveal any temporal changes, possibly due to the fewer sampling months included in that study [31].

Consistent with the NMDS analysis based on OTU abundances, the taxonomic analysis also revealed a clear distinction between the gill microbiome of the mussel *M. galloprovincialis* and the communities in the surrounding seawater and sediment. The high-rank microbial taxa detected in the surrounding seawater, as well as the higher proportion of *Thermoplasmatota* Marine Group II

during the colder period of the year, are typical of seawater microbial communities in the Adriatic Sea [100, 101]. Similarly, the high-rank microbial taxa detected in the surrounding sediment have previously been found in surface sediments of the Adriatic Sea [50, 102]. At higher taxonomic ranks, the distinction between the gill microbiome and the microbial communities in the surrounding seawater and sediment was mainly due to the high proportion of *Gammaproteobacteria* in the gill microbiome. The high proportion of taxa belonging to this class in the gill microbiome has also been observed in other studies focusing on the gills of the mussel *M. galloprovincialis* [14, 31]. Because of the high proportion of *Gammaproteobacteria* in the gill microbiome, no differences between the two locations, Lim Bay and Mali Ston Bay, or between sampling months were observed. However, when the analysis was performed at lower taxonomic ranks, such as family and genus, taxa were identified that were either resident, based on their consistent presence at both locations and throughout the study period, or specific to certain locations or months, likely exhibiting transient associations with the mussel host. In addition, samples that did not yield sufficient sequences for further analysis were also specific to a particular month, namely January 2021. This peculiarity may be attributed to lower microbial counts in the gills in winter, as many microbes exhibit reduced growth at lower water temperatures, regardless of the bivalve host species or tissue [7, 103–105].

Although members without known relatives within *Endozoicomonadaceae* showed high variability in their proportion within the gill microbiome of *M. galloprovincialis* between individuals, they were detected in all analysed mussels, showed minimal differences in proportion between mussels from different locations or months, and were only sporadically detected at very low proportions in seawater. This suggests that this group may be regarded as putatively resident in the gills of *M. galloprovincialis*. It is important to note that, although consistent detection of members of this group across a wider geographic area suggests a strong association with this host [8], applying methodologies to multiple bivalve host species that allow strain tracking, as well as approaches enabling the localisation of microbial cells in tissues, such as fluorescence in situ hybridisation (FISH), is crucial for verifying their residency [106–108]. The family *Endozoicomonadaceae* is known to form associations with many marine benthic invertebrates, including sponges and

bivalves, as well as with some fishes [6, 106, 109, 110]. Furthermore, members of this family have
 previously been detected in the gills of *M. galloprovincialis* [111], as well as in the gills of *Mytilus*
edulis [110] and *Mytilus californianus* [112], two bivalve species related to *M. galloprovincialis*.
 Genomic analyses suggest that members of this family can rapidly adapt and transition between
 free-living and host-associated lifestyles, resulting in host–microbe relationships that can range from
 mutualistic to parasitic depending on the host species and ecological context [106]. Although they
 are widely recognised as beneficial symbionts involved in host health, disease, and stress tolerance,
 as well as in multiple other host functions such as regulating nutrient provision, nitrogen and sulphur
 cycling, metabolism, and microbiome stability [106, 108, 113], some studies have linked them to
 pathogenic roles in corals [114] and fish [115, 116]. Their dual role in marine organisms is also
 evident in bivalve hosts, where *Endozoicomonadaceae* often dominate the microbial community
 supporting nitrogen assimilation [117], digestion, and immunomodulation [6, 118], but have also
 been associated with poor physical condition [119], pathology, and localised mortality events
 [111, 120]. For example, Cano et al. [111] reported that members of this family may form
 intracellular colonies or large extracellular aggregates in the gills and other tissues of various
 bivalve species, including *Mytilus* spp., which were occasionally associated with tissue lesions of
 varying intensity. While the exact functional role of *Endozoicomonadaceae* in *M. galloprovincialis*
 remains unknown, the presence of this group in all analysed mussels in the present study suggests a
 mutualistic interaction. Nevertheless, although no external symptoms of pathology were recorded
 in populations tested during this study and a putatively beneficial role can be hypothesised, the
 potential for an *Endozoicomonadaceae* infection to compromise *M. galloprovincialis* tissue integrity
 and overall fitness cannot be excluded and warrants further study. It is also important to note that
 this group consists of sequences without currently known relatives (i.e. unclassified sequences) that
 likely represent poorly characterised lineages. Improved taxonomic resolution and identification of
 novel taxa are therefore required to provide better insight into the temporal and spatial dynamics of
 taxa within *Endozoicomonadaceae* and to determine their interactions with this host species.

In the present study, *Endozoicomonas*, the type genus [121] and the only identified genus

within the *Endozoicomonadaceae* family occurring at higher proportions, showed a distribution distinct from that observed for unclassified sequences within this family. Although *Endozoicomonas* has previously been identified as a component of the gill microbiome of *M. edulis* [110], the results of this study, which accounted for potential temporal fluctuations over one year, revealed its near absence from the gill microbiome at both locations from July to November 2020, when seawater temperatures at both sites were above 17 °C [45]. However, from January to May 2021, corresponding to the period from winter to spring, when temperatures ranged from 11 to 16 °C [45], it comprised a substantial proportion of the microbiome in many individuals from both locations, indicating the transient nature of this taxon and possibly a different host–microbe interaction. The temporal pattern of *Endozoicomonas* observed in this study is broadly analogous to that reported for *Endozoicomonas* in *M. galloprovincialis* from the Atlantic coast of Spain [6]. It appears that the proportion of *Endozoicomonas* in the gill microbiome of *M. galloprovincialis* from the Adriatic Sea may be linked to lower temperatures during the colder season. Studies in corals suggest that heat stress may lead to a decline in the abundance and/or compositional shifts of *Endozoicomonas*, depending on the host species [122]. Conversely, some specific *Endozoicomonas* strains from the gill tissue of the invasive oyster *Spondylus spinosus* appear to be adapted to high temperatures, not thriving below winter temperatures of 11 °C [123]. Nevertheless, in these previous studies, a decline in *Endozoicomonas* abundance has often been associated with a concomitant increase in the proportion of opportunistic bacteria or pathogens, which largely agrees with the results of this study. Notably, the increased proportion of *Endozoicomonas* in the gills of *M. galloprovincialis* reported here coincides with the late-winter and early-spring peak in gametogenesis and spawning activity of this bivalve species in the Adriatic Sea [45]. As *Endozoicomonas* genomes are enriched in genes associated with carbohydrate and protein metabolism [124], their presence may be linked to the high energy demands during the reproductive phase. However, further studies are needed to elucidate definitively the specific contribution, temporal dynamics, and functional mechanisms of *Endozoicomonas* in the gill microbiome of *M. galloprovincialis*.

“*Candidatus* Endoecteinascidia” comprised a substantial proportion of the gill microbiome

in many mussels from both locations in July 2020, while in other months, only a few individuals' microbiomes contained a higher proportion of this group, indicating its transient nature. This group has been described as an endosymbiont of the ascidian *Ecteinascidia turbinata* [125]. Interestingly, genome analysis revealed an extreme reduction of the genome in this group and the presence of biosynthetic genes required for the production of ecteinascidin 743 (ET-743), a chemotherapeutic natural product thought to be pivotal for the interaction with this tunicate host [126–128]. These data suggest that members of this group form a mutualistic relationship with the host, so at present it is difficult to explain the observed transient nature of “*Candidatus* Endoecteinascidia” in the gill microbiome of *M. galloprovincialis* observed in this study. It is possible that the host of this group is not the mussel but an ascidian biofouling species, and its detection is due to the proximity of the ascidians to the mussel gills. Additionally, this group may be taken up by the mussel as particles from decaying ascidians because, as mentioned previously, particle-attached microbes are more likely to be retained by the gills [39, 97, 98]. Ascidians are important members of the biofouling community on various aquaculture bivalve species, including *M. galloprovincialis* [129], and their presence has also been observed in aquaculture farms in the northern Adriatic Sea, including Lim Bay [130]. In addition, ascidians in the Mediterranean Sea, including the genus *Ecteinascidia*, have shown marked seasonality of growth on artificial plates deployed next to a bivalve aquaculture facility [131], which could explain the observed higher proportion of its endosymbiont “*Candidatus* Endoecteinascidia” in the gill microbiome during the summer peak in July 2020 compared to other sampling months.

Groups that were transient in nature and also differed between the two aquaculture locations, Lim Bay and Mali Ston Bay, both belonged to the family *Vibrionaceae* and included *Vibrio* as well as members without known relatives within *Vibrionaceae*. Members of the family *Vibrionaceae* are often found in the microbiome of various bivalve species, including *M. galloprovincialis* [6, 14, 16, 21, 23], forming relationships with their bivalve hosts that can range from mutualistic to neutral or parasitic [104]. In addition to their preferential free-living lifestyle, members of the *Vibrio* community can also exhibit a particle-associated lifestyle, increasing the likelihood of being

582 captured by the gills [39, 97, 98, 132]. In this study, a preference of *Vibrio* and members without
583 known relatives within *Vibrionaceae*, i.e. sequences that could not be classified to any genus within
584 *Vibrionaceae*, for the period from September to November 2020, as well as for the aquaculture
585 location in Lim Bay, was observed. The higher proportion of *Vibrionaceae* observed during this
586 period is consistent with a study using culture-dependent methods focusing on the gills of *M.*
587 *galloprovincialis*, sampled during the same period and at the same aquaculture locations, which
588 revealed a positive correlation between total *Vibrio* counts and temperature (L. Perić, personal
589 communication). This is not surprising, as many species of *Vibrio* are known to thrive at higher
590 temperatures [104, 105, 132]. The observed difference between Lim Bay and Mali Ston Bay could
591 be due to distinct hydrological conditions and productivity at these locations, resulting in greater
592 accumulation of organic matter and microbial growth in Lim Bay [41–44]. Indeed, all described
593 *Vibrio* spp. are obligate heterotrophs that rely on organic matter as a carbon source [133]. Finally,
594 improved taxonomic resolution and identification of novel taxa within *Vibrionaceae* would provide
595 better insight into the temporal and spatial dynamics of members of this family and reduce the
596 proportion of unclassified *Vibrionaceae* sequences observed in this study.

Conclusions

Because studies assessing the microbiome of bivalve species, including the mussel *M. galloprovincialis*, have focused on gut tissue [2, 11, 16, 21, 28, 30], knowledge of the temporal and spatial dynamics of microbiomes from other body parts of this important aquaculture species remains limited. This study analysed a comprehensive set of gill microbiomes and compared them with microbial communities in the surrounding seawater and sediment to address this gap. Overall, the structure of the gill microbiome of *M. galloprovincialis* at the OTU level was distinct from the microbial communities in the surrounding seawater and sediment, and did not differ between the aquaculture locations, Lim Bay and Mali Ston Bay. However, the structure of the gill microbiome varied between sampling months, suggesting temporal dynamics. Furthermore, taxonomic analysis revealed the presence of both resident and transient microbial taxa in the gill microbiome. Members without known relatives within *Endozoicomonadaceae* (i.e. unclassified *Endozoicomonadaceae*) were observed in the gill microbiome of mussels from both locations and throughout the study period, indicating their resident nature. In contrast, *Endozoicomonas* and “*Candidatus* Endoecteinascidia” were transient, appearing in the gill microbiome of mussels from both locations only in certain months, as were *Vibrio* and members without known relatives within *Vibrionaceae* (i.e. unclassified *Vibrionaceae*), with the distinction that these were also location-specific, being characteristic of Lim Bay. It is worth noting that, in this study, differences in taxon proportions were determined and tested, limiting the assessment of differences in absolute abundances due to the compositional nature of the 16S rRNA amplicon data. Finally, this study emphasises that multiple biological replicates are essential for capturing inter-individual variation in bivalve microbial profiles, underscoring the importance of the results obtained. Thus, our findings provide a robust baseline that substantially contributes to ongoing research into tissue-specific marine bivalve host–microbiome interactions.

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Author contributions

MK and LP conceived and designed the study. LP managed the project and secured funding. DK, IVS, KO, VK, JB, SBČ, and VN conducted fieldwork and collected samples. MK and MM processed the samples and performed molecular analyses. MK conducted the bioinformatic analyses and statistical tests, and wrote the first draft of the manuscript. All authors reviewed and approved the final manuscript.

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Data availability

Nucleotide sequences of the V4 region of the 16S rRNA gene obtained in this study have been deposited in the European Nucleotide Archive at EMBL-EBI under accession number PRJEB107277 (<https://www.ebi.ac.uk/ena/browser/view/PRJEB107277>). The detailed analysis procedure, including the R Markdown file for this article, is available on GitHub (https://github.com/MicrobesRovinj/Korlevic_MusselMicrobiome_MicrobEcol_2026) and Zenodo (<https://doi.org/10.5281/zenodo.21790743>).

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